

ARITHRA MAYUR

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SUMMARY

Data Science undergraduate building production ML and GenAI systems — LLM fine-tuning, retrieval-based pipelines, and backend APIs — with end-to-end delivery from model development to containerized deployment and CI/CD.

EDUCATION

Institute of Advanced Education and Research

B.Sc. in Data Science — CGPA: 8.64/10

Rajarhat, WB

2024 – 2028

EXPERIENCE

Backend AI Engineering Intern

July 2026 – Present

FlyRank

- Built FastAPI REST APIs with SQLite persistence and a layered repository architecture for CRUD operations.
- Implemented Supabase Auth and JWT-based authentication with protected routes and token verification.
- Developed a background job system with queue processing, worker execution, and job status tracking.

ML Intern

June 2025 – July 2025

InternPe

- Expanded a movie rating prediction project into an end-to-end ML system using a stacking ensemble (XGBoost + LightGBM + Ridge), improving R^2 by approximately 32% through Optuna tuning (verify exact baseline before quoting).
- Engineered 34 tabular and NLP sentiment features using VADER and DistilBERT, integrating SHAP for model interpretation.
- Developed a FastAPI API and Streamlit dashboard, containerized with Docker, automated CI/CD using GitHub Actions, and tracked experiments with MLflow.

PROJECTS

DomainLens | Python, Transformers, PEFT, TRL

[GitHub](#)

- Fine-tuned a legal QA model on 820 human-verified QA pairs from 300 Supreme Court judgments, improving ROUGE-1 F1 by approximately 49%.
- Built a failure-analysis pipeline identifying hallucination and reasoning errors, reducing wrong-reasoning errors by 9.5pp through targeted model correction.
- Evaluated reliability on 100 GSM8K problems, maintaining 37% accuracy, and deployed an interactive Streamlit legal QA application.

Competitor Intelligence Agent | Python, FastAPI, Groq LLM, ChromaDB

[GitHub](#)

- Built an autonomous competitive intelligence platform monitoring 4 data sources and classifying 11 competitor change types using Groq LLM, Sentence Transformers, and ChromaDB for retrieval.
- Designed confidence scoring that reduced roughly 80% of LLM API calls through semantic similarity filtering and structured evidence generation.
- Implemented adaptive source reliability learning with automated sales battlecards and analyst briefings delivered in under 5 minutes.

ML-Powered Cyber Threat Prioritization | SentenceTransformers, KMeans, MLflow, Docker

[GitHub](#)

- Built an ML-based threat ranking system achieving 90.7% precision and 65.7% recall@150, improving baseline ranking by approximately 108%.
- Implemented semantic similarity and explainable risk scoring using SentenceTransformers and MITRE ATT&CK, clustering 405 threats into 10 attack patterns.
- Deployed the pipeline with MLflow tracking, Docker, pytest (82% coverage), and GitHub Actions CI/CD.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++

ML & Data: Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Optuna

Deep Learning & NLP: PyTorch, Transformers, PEFT, TRL, DistilBERT, Sentence Transformers

LLMs & Generative AI: LLM Fine-Tuning, DPO, RAG, ChromaDB, Groq API

MLOps & Tools: FastAPI, Flask, Docker, MLflow, Git, GitHub Actions, Streamlit, SQLite

CODING PROFILES

LeetCode: 100+ problems solved across SQL, Data Structures & Algorithms, and Pandas

HackerRank: Python & SQL badges

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